

GREAT LAKES INSTITUTE OF MANAGEMENT, GURGAON

Post Graduate Diploma in Management | PGDM 2025–27

SUMMER INTERNSHIP PROJECT REPORT

PROJECT TITLES

- 1) **Optimizing Inventory Flow of Die-Cast Automotive Components using Cycle Time Analysis and Bottleneck Identification approach**
- 2) **Daily breakdown analysis and development of an OEE Tracking and Monitoring Model for GDC and LPDC Machines**

SUBMITTED BY

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Spark Minda

Industry / Sector

Auto Component Manufacturing

Location

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Company Guide Name & Designation

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Designation

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Department

Operations

SUBMISSION DETAILS

20/06/2026

Submitted in partial fulfilment of the requirements of the Post Graduate Diploma in Management

Great Lakes Institute of Management, Gurgaon

Date: 04th June 2026

Subject – Internship Completion Certificate

To Baidurja Biswas,

This is to certify that **Mr. Baidurja Biswas**, a student of **Post Graduate Diploma in Management from Great Lakes Institute of Management, Gurgaon**, has successfully completed his designated Summer Internship with Spark Minda Die Casting Division II, Pune from **1st April 2026 to 4th June 2026**.

During his internship, he was associated with the Production Planning and Control Department, where he primarily focused on Inventory planning Projects and optimization of existing datasets. Throughout the internship period, Baidurja displayed a sincere attitude, professionalism, and an eagerness to learn, which contributed positively to the department's ongoing initiatives.

We appreciate his efforts and contributions during the internship tenure and wish him the very best for his future endeavours and professional journey ahead.

For MINDA CORPORATION LTD, (Die Casting Div.)



Ranjeet Patil

Lead – HRBP & IR



MINDA CORPORATION LIMITED (DIE CASTING DIVISION - II)

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Registered Office: A-15,Ashok Vihar, Phase-I, Delhi - 110052.

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CERTIFICATE

It is certified that Mr. Baidurja Biswas, a PG student of Great Lakes Institute of Management, Gurgaon, has undergone a Summer Internship at Spark Minda Die Cast Division Plant II from 01/04/2026 to 04/06/2026

His/ her project titled "Designing an optimized inventory flow for selected die cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification" and "Developing an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines" has been completed to our satisfaction and the present report is a bona fide account of the same.



Milind Sarawade
Manager PPC & Schedules

Signature, Name, and designation
and designation of the Industry Guide

Date: 04/06/26

Signature and Name
of the Institute Guide

Date:

OVERVIEW

Name – Baidurja Biswas

Institute Name – Great Lakes Institute of Management, Gurgaon

Organization Name – Spark Minda

Introduction to the Company

Spark Minda was started in 1958 by Shadi Lal Minda and operates in the OEM for automobile companies. Came into the B2C segment in 2022 by launching “Minda Helmet”. The company has 34 plants and offices, of which 14 are in North India, 13 in West India, 4 in South India, 3 in ASEAN countries and 1 in East Asia.

Major equipment manufactured by Spark Minda includes car door locksets, die-cast components, starter motors, instrument clusters, Sensors, wiring harnesses, EV components, etc.

Some of the major die-cast products made by Spark Minda are

T6 Lp Turbo Charger used by BMW and Audi

1.5 Litre i-DTEC CH used in Honda Amaze

CH-1.5L Diesel used in Ford Ecosport

CH 1.0L Diesel used in Volvo MDS Bus

Role – Operations Intern

Duration of SIP – 9 weeks

Industry of the Company – Auto Component Manufacturing

Name of Mentor in College and Company

College Mentor– Dr. Surya Prakash

Organization Mentor- Mr. Milindkumar Shivaji Sarawade, Senior Manager, Production Planning and Control (PPC) Department, Die Cast Division, Plant II, Spark Minda, Pune, Maharashtra

Project Allocated

1. Optimizing Inventory Flow of Die-Cast Automotive Components using Cycle Time Analysis and Bottleneck Identification approach
2. Daily breakdown analysis and development of an OEE Tracking and Monitoring Model for GDC and LPDC Machines

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Spark Minda, Die Cast Division, Pune, for providing me with the opportunity to undertake this internship and gain firsthand exposure to real-world operations in a world-class automotive manufacturing environment.

I am deeply grateful to my industry mentor, Mr. Milindkumar Shivaji Sarawade, Senior Manager, PPC department, for his invaluable guidance, patience and willingness to share his domain expertise throughout the duration of this programme. His insights into shop floor operations, production planning and constraint management were instrumental in shaping the direction and quality of this work

I would also like to thank all the operators, supervisors and staff at the Plant for their cooperation during data collection and for taking the time to explain on-ground processes with clarity and openness.

I extend my gratitude to the faculty and administration of Great Lakes Institute of Management, Gurgaon, for their academic support and for helping me get this internship opportunity.

I am thankful to my parents and all my peers for their constant encouragement and support throughout this journey.

EXECUTIVE SUMMARY

During the Summer internship at the Spark Minda Die Cast Division Plant in Pune, two projects were undertaken to improve shop-floor efficiency in a high-volume automotive component manufacturing environment. The first project focused on designing and developing an optimized inventory flow for four die-cast components: Porsche housing RH, Phinia 6-cylinder, Borgwarner EAD HP, and LP by conducting cycle time analysis, lead time analysis, and capacity utilization-based bottleneck identification. A Drum buffer rope scheduling model, along with stagewise safety stock, was used to recommend a synchronized production flow, control WIP buildup, and protect against machine breakdowns. The second project involves developing a structured OEE (Overall Equipment Effectiveness) tracking and monitoring system for both GDC and LPDC machine lines by consolidating daily breakdown and efficiency data, performing data cleaning, and building a weekly performance dashboard in Excel that enabled plant leadership to identify the breakdown points and help in data-driven production planning. Together, both projects delivered an improved operational visibility and laid the foundation of a leaner and more optimized production system capable of consistently meeting client demand.

CHAPTER 1

PROJECT OVERVIEW

PROJECT 1

1.1 Objective

Designing an optimized inventory flow for selected die cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification

1.2 Scope of the Project

- Studying the end-to-end production flow of the die cast components, Phinnia 6 cylinder and Porsche housing RH
- Measuring and analyzing the cycle time of all the stages involved in the production process
- Identifying the bottlenecks and constraints affecting the process flow
- Estimating safety stock for different stages to minimize production interruption with appropriate assumptions.

1.3 Expected Outcome

- Identification of critical bottlenecks affecting the production process
- Better inventory management through better safety stock planning
- Enhanced production continuity and material flow across operations

PROJECT 2

1.4 Objective

Developing an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines

1.5 Scope of the project

- Collecting and studying the daily operations data of GDC and LPDC machines
- Verifying the data and doing data cleaning using appropriate methods
- Developing a systematic OEE tracking and monitoring framework for continuous machine performance evaluation
- Studying the machine utilization patterns and detecting the key breakdown points

1.6 Expected Outcome of the Project

- A structured OEE measurement and monitoring sheet
- Improved visibility into machine efficiency and breakdown points
- Identification of crucial breakdown points and downtime issues

CHAPTER 2

LITERATURE REVIEW

2.1 OBJECTIVE

To determine and evaluate the main production system bottleneck and apply Theory of Constraints-based techniques, particularly Drum-Buffer-Rope scheduling, to increase system throughput and lower Work-In-Progress buildup in the production process.

2.2 INTRODUCTION

In today's manufacturing world, production efficiency is essential to achieving and maintaining a competitive advantage. Bottlenecks and inventory management are two key areas of focus that can play an important role in improving production efficiency. A bottleneck is a point in the system where the output of the entire process is limited, and therefore must be managed effectively for maximum production efficiency (Goldratt and Cox, 1984). Inventory that is poorly managed, particularly work-in-progress (WIP) inventory, can hide the true extent of inefficiencies in a system and increase costs (Hopp and Spearman, 2008). This study reviews the literature on bottleneck management, Theory of Constraints (TOC), flow of inventory and Lean manufacturing concepts, to provide a framework for understanding how to increase throughput in real-world manufacturing operations.

2.3 BOTTLENECK IN MANUFACTURING

Bottleneck resource refers to a resource's ability to provide sufficient output to meet the demands placed on it at any given time. Thus, the available throughput of a system will be limited by whichever bottleneck resource exists. The study (Goldratt & Cox, 1984) determine that some of the main causes of bottlenecks are due to specific issues related to resources, namely: machine failures/breakdowns, variability in setup time, operator skill gaps, and unbalanced workloads, among others. Further contributing to establishing new constraints are both demand variability and poor batch sizes (Terwiesch et al., 2001).

2.4 BOTTLENECK IDENTIFICATION

Methods for identifying bottlenecks generally fit into three categories: Observational methods, Utilization methods and Simulation methods (Roser et al., 2002). Utilization methods (where the machine with the highest average utilization is marked as a bottleneck) are commonly used because these are easy to implement (Kuo et al., 1996), but these can be prone to inaccuracies due to high variability. The Active Period method (Roser et al., 2002) should be considered as a better alternative to the Utilization method, as it locates which machine has the most frequent occurrences of blocking or starving other machines.

It is important to differentiate between static and dynamic bottlenecks. Static bottlenecks remain a bottleneck on only one workstation. Dynamic bottlenecks will move (or migrate) in response to changes in machine uptime, demand fluctuations, and changes in product mixes (Li et al., 2009). Recognizing this distinction is very important when working with real-world manufacturing environments, for they too may move as fast as a single shift of production.

2.5 THEORY OF CONSTRAINTS

According to Goldratt & Cox (1984), in The Theory of Constraints (TOC), there is a minimum of one constraint that limits system performance. Improving anything else other than the constraint will not yield meaningful throughput improvement. Manufacturing management is now approached through TOC by focusing on optimizing local efficiencies, using all decisions to maximize the throughput of the constraint.

2.6 RELEVANCE TO THE STUDY

While there is a great deal of theoretical progress that has already taken place in the area of bottleneck detection, scheduling using the Theory of Constraints (TOC), and Lean improvements, there is still a significant gap between theory and practice. Most literature on bottlenecks utilizes a theoretical framework or simulates data in controlled environments (Roser et al., 2002; Li et al., 2009). The actual implementation of a bottleneck or TOC schedule in the real world, especially for applications involving high levels of variability in demand, scheduling products that are not the same across all manufacturing processes, and taking into account human factors associated with manufacturing, is still relatively unexplored in the literature (Rahman, 1998; Watson et al., 2007).

CHAPTER 3

PROJECT 1

3.1 OBJECTIVE

To design an efficient and optimized inventory flow for selected die cast components of Phinia 6 Cylinder Porsche Housing RH and Borgwarner EAD HP and EAD LP by analyzing machine cycle times, identifying production bottlenecks and applying appropriate concepts to improve process efficiency and material flow.

3.2 METHODOLOGY AND DATA COLLECTION

- Visited the Plant and observed the end-to-end manufacturing process of the products, Phinia 6 Cylinder, Porsche housing RH and Borgwarner EAD LP and EAD HP, and noted down the cycle times of the different machines involved in the entire process.
- From the data, the bottlenecks were identified, and based on the monthly demand plan, an inventory flow was made accordingly, considering the safety stock and buffer stock assumptions and applying appropriate concepts of Theory of Constraints (TOC).

The Cycle times of the Processes and the different sequential stages involved in production are given below

Porsche Housing RH

Table 3.1 Cycle time information of Porsche housing RH

Step	Cycle time
Melting	1 ton 1 hr
Coring	220 sec
GDC (Stationary)	370 sec
Gate cutting 1	60 sec
Decoring 1	55.5 sec
T5	8 hr for 90 pcs
Decoring 2	55.5sec
Gate cutting 2	40sec
T6	104 pcs 12 hrs, we can add implicitly every 2 hrs
Fittling + Lensing	160 sec
Shot blasting	5 mins 40 units
Quality Inspection	120 sec
CNC 1	175 sec
CNC 2	196 sec

VEMC	151 sec
Final Inspection	90 sec
Washing	90 sec 2 pcs together
Leak test	60 sec
Screw Assembly	60 sec
Multigaging	60 sec
CSL	90 sec
Packing	60 sec

3.3 PROCESS FLOW

Studied the end-to-end process flow of Porsche housing RH and monitored the steps involved in the production process to make an inventory plan.

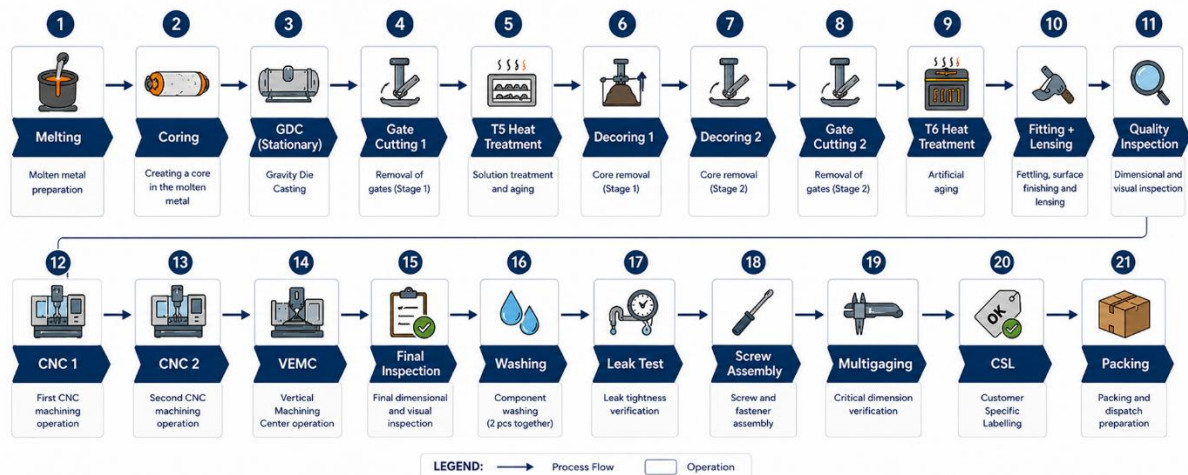


Fig 3.1 Production process flow of Porsche housing RH

3.4 LEAD TIME ANALYSIS

A lead time analysis was conducted for the Porsche housing RH manufacturing line to understand the material flow from initial melting to the final finished good. The analysis was done under initial production startup conditions, assuming no WIP is present in stages. The analysis aimed to identify the bottlenecks, the accumulation delays and their impact on overall lead time.

Once steady-state production is achieved, material flow becomes continuous, and downstream operations receive regular inputs from the upstream process.

The cumulative lead time was calculated by analyzing the cycle time, batch size and accumulation delays between the processes.

Table 3.2 Lead time analysis of Porsche Housing RH

Steps	Cycle time	Explanation	Leadtime (hrs)
Melting	1 ton 1 hr		1
Coring	220 sec	3 machines are available	1.36
GDC (Stationary)	370 sec	18 casts are produced per hour and sent continuously for Gate cutting	6.36
Gate cutting 1	60 sec	18 casts cut in 18mins	6.66
Decoring	55.5 sec	18 casts decoring in 16 mins	6.92
T5	8 hr 90 pcs	We have to wait here for 4hrs accumulation of atleast 90 pcs to proceed with T5. Every 1.56 hour we get 18 pcs	18.92
Decoring	55.5sec	17 casts decoring in 16mins	19.18
Gate cutting 2	40sec	17 casts decoring in 11mins	19.36
T6	104 pcs 12 hrs we can add implicitly every 2 hrs	This process is semi-continuous it can produce 104 product in 12 hrs but these can be added impartially in every 2 hr period so we divide 12 hrs into 6 intervals and as capacity is 104 we can add 17 units in every interval . 1 unit is kept as WIP	31.36
Fittling + Lensing	160 sec	17 units fittling lensing in 45.3 mins	32.11
Shot blasting	5 mins 40 units	We have to again wait here for accumulation of 40 units we get 17 pcs out in 27 hrs approx and every 2hrs we will get 17 units so we can start with this step after 5.5 hrs of waiting with the first lot of 17 units	34.94
Quaity Inspection	120 sec	For 40 units it will 80 mins to complete quality inspection	36.24
CNC 1	175 sec	CNC 1 of 40 units will be done in 0.9 hrs	37.14
CNC 2	196 sec	CNC 2 of 40 units will be done in 1 hr approx	38.14
VEMC	151 sec	VEMC of 40 units will be done in 1.67 hrs	39.81
Final Inspection	90 sec	Inspection of 40 units will be done in 1 hrs	40.81
Washing	90 sec 2 pcs	Washing of 40 units done in 0.5 hrs	41.31
Leak test	60 sec	Leak test will be done in 40 mins	41.91
Screw Assembly	60 sec	Screw Assembly in 40 mins	42.51
Multigaging	60 sec	Multigaging in 40 mins	43.11
CSL	90 sec	CSL will be done 1 hr	44.11
Packing	60 sec	Packing will be done in 40 mins	44.71

From this, it was found that T6 is the main bottleneck of the production flow of Porsche housing RH production, so we have to follow the drum buffer rope model to make the production flow as per the capacity of T6 in the process.

3.5 SAFETY STOCK CALCULATION AND INVENTORY PLANNING

The monthly demand for Porsche RH from the client is 5000 units

The number of working days in the plant is 26 days

Total hours, including all shifts per day, is 24 hrs and 86400 secs

Based on this, the daily production is 192 units per day (5000/26)

Hourly Production rate is 8 units/hr (192/24)

TAKT time is 449.28 secs (86400/192)

The safety stock factor we have considered for this was 1.5

The safety stock formula used is

Minimum Safety stock = Recovery time * Safety stock factor * Demand rate per hour

To determine recovery time, we calculated the machines' capacity utilization. The capacity utilization is based on the daily production demand of 192 units/day and does not include safety stock processing. This assumption is intentional because if capacity utilization were calculated including safety stock, it would overstate the load, especially when the production reaches a steady state, and distort the recovery time classification for the machines.

The table shows the amount of Safety stock to be kept in each stage so that machine breakdown or other delays don't stop the production process, and the process is steady and continues uninterrupted.

Recovery Time Assumption

For machines with more than 20% capacity utilization, we have considered a 2hr recovery time.

For machines with less than 20% capacity utilization, we have considered a 1 hr recover as these are fast and can cope with the process delays.

And for the batch process, we will keep one batch and a little extra, considering the delay as a safety stock

Table 3.3 Safety Stock analysis of Porsche Housing RH

Step	Cycle time	Capacity	Per day produce	Parts in WIP	Parts in safety stock	Capacity Utilization	Minimum Safety Stock
Melting	1 ton 1 hr						20
Coring	220 sec	1178	193			16%	12
GDC (Stationary)	370 sec	432	193	193		45%	24
Gate cutting 1	60 sec	1296	193			15%	12
Decoring	55.5 sec	1401	193			14%	12
T5	8 hr 90 pcs	243	193	90	103	Batch	100
Decoring	55.5sec	1401	193			14%	12
Gate cutting 2	40sec	1944	193			10%	12
T6	104 pcs 12 hrs we can add implicitly every 2 hrs	208	193			93%	24
Fittling + Lensing	160 sec	486	193			40%	24
Shot blasting	5 mins 40 units	10368	193			2%	12
Quaity Inspection	120 sec	648	193			30%	24
CNC 1	175 sec	444	193			43%	24
CNC 2	196 sec	397	193			49%	24
VEMC	151 sec	515	193			37%	24
Final Inspection	90 sec	864	193			22%	24
Washing	90 sec 2 pcs together	1728	193			11%	12
Leak test	60 sec	1296	193			15%	12
Screw Assembly	60 sec	1296	193			15%	12
Multigaging	60 sec	1296	193			15%	12
CSL	90 sec	864	193			22%	24
Packing	60 sec	1296	193			15%	
Total WIP							456

Phinia 6 Cylinder

Table 3.4 Cycle time information of Phinia 6 cylinder

Step	Cycle time
Melting	1 ton in 1 hour
GDC (Tilting)	514.3 secs. We have 4 machines one can produce 7 castings in an hour
Gate Cutting	80 sec for 2 pcs
Heat treatment	12 hr for 104 pcs
Fitling + Lensing	120 secs
Shot blasting	Batch of 50 in 12 mins
Quality Inspection	120 secs
Preturning	328 secs
OP 10	700 secs
Washing	80 secs
CNC	40 secs
Air gauge	120 secs
Pump housing bush press	300 sec
OP 20	664 secs
OP 30	690 secs
Washing 1	60 sec for 2 pcs together
ECM	72 secs
Washing 2	30 secs
LT	210 sec
Multi	270 sec
Endoscopy	180 secs
CSL	300 secs
Ultrasonic Washing	10 units 18 mins
Packing	60 sec

3.6 PROCESS FLOW

Studied the end-to-end process flow of Phinia 6 cylinder and monitored the steps involved in the production process to make an inventory plan.

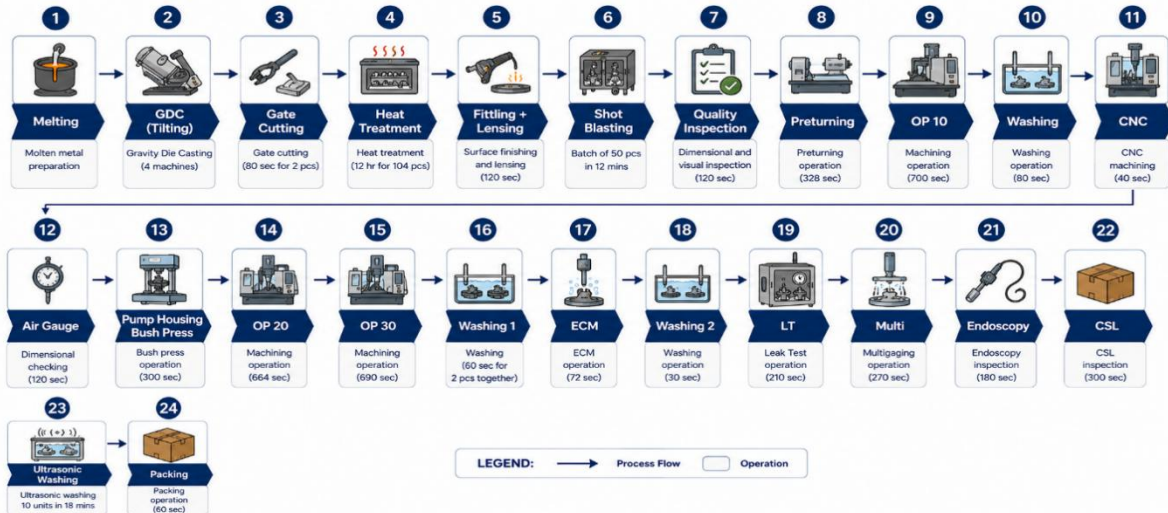


Fig 3.2 Production process flow of Phinia 6 Cylinder

3.7 LEAD TIME ANALYSIS

A lead time analysis was conducted for the Phinia 6-cylinder manufacturing line to understand the material flow from initial melting to the final finished good. The analysis was done under initial production startup conditions, assuming no WIP is present in stages. The analysis aimed to identify the bottlenecks, the accumulation delays and their impact on overall lead time.

Once steady-state production is achieved, material flow becomes continuous, and downstream operations receive regular inputs from the upstream process.

The cumulative lead time was calculated by analyzing the cycle time, batch size and accumulation delays between the processes.

Table 3.5 Lead Time Analysis of Phinia 6 Cylinder

Step	Cycle time	Explanation	Lead Time
Melting	1 ton in 1 hour		1
GDC (Tilting)	514.3 secs. We have 4 machines one can produce 7 castings in an hour	It will produce 25 castings in 0.5 hours and send them for the subsequent process. It takes approx 4hr for cooling	5.8
Gate Cutting	80 sec for 2 pcs	25 castings are cut in 16.6 mins	6.1
Heat treatment	12 hr for 104 pcs	25 are kept and we have to wait for 2.4hr to prepare a batch of 100 pcs for batch	21.5
Fitling + Lensing	120 secs	50 pcs will be done in 100 mins and sent to the next process	23.1
Shot blasting	Batch of 50 in 12 mins	50 pcs will be done in 12 mins	23.3
Quality Inspection	120 secs	50 units will be done in 100 mins	24.9
Preturning	328 secs	The 50 units will be sent continuously through all the stages	24.99
OP 10	700 secs	We have two lines available so it can handle the required units	25.18
Washing	80 secs	All the 50units will continue through the machine shop processes in a sequential manner	25.2
CNC	40 secs		25.21
Air gauge	120 secs		25.24
Pump housing bush press	300 sec		25.32
OP 20	664 secs		25.5
OP 30	690 secs		25.69
Washing 1	60 sec for 2 pcs		25.8
ECM	72 secs		25.82
Washing 2	30 secs		25.85
LT	210 sec		25.9
Multi	270 sec		25.97
Endoscopy	180 secs		26.02
CSL	300 secs		26.1
Ultrasonic Washing	10 units 18 mins	10units get processed in batch the other products wait	28.2
Packing	60 sec		29.8

From the analysis, the primary bottleneck is OP 10, followed by OP 20 and OP 30. However, since parallel production lines are available for all three, it doesn't create a significant impact on the production process.

3.8 SAFETY STOCK CALCULATION AND INVENTORY PLANNING

The monthly demand for Phinia 6 cylinder from the client is 3600 units

The number of working days in the plant is 26 days

Total hours, including all shifts per day, is 24 hrs and 86400 secs

Based on this, the daily production is 138 units per day (3600/26)

Hourly Production rate is 6 units/hr (138/24)

TAKT time is 624 secs (86400/138)

The safety stock factor we have considered for this was 1.5

The safety stock formula used is

Minimum Safety stock = Recovery time * Safety stock factor * Demand rate per hour

To determine recovery time, calculated the machines' capacity utilization. The capacity utilization is based on the daily production demand of 192 units/day and does not include safety stock processing. This assumption is intentional because if capacity utilization were calculated including safety stock, it would overstate the load, especially when the production reaches a steady state, and distort the recovery time classification for the machines.

The table shows the amount of Safety stock to be kept in each stage so that machine breakdown or other delays don't stop the production process, and the process is steady and continues uninterrupted.

Recovery Time Assumption

For machines with more than 20% capacity utilization, we have considered a 2hr recovery time.

For machines with less than 20% capacity utilization, we have considered a 1 hr recover as these are fast and can cope with the process delays.

And for the batch process, we will keep one batch and a little extra, considering the delay as a safety stock.

Table 3.6 Safety stock analysis of Phinia 6 cylinder

Step	Cycle time	Capacity	Per day produce	Parts in WIP	Parts in safety stock	Capacity utilisation	Minimum Safety stock
Melting	1 ton in 1 hour						30
GDC (Tilting)	514.3 secs. We have 4 machines one can produce 7 castings in an hour	151	139	139		92%	18
Gate Cutting	80 sec for 2 pcs	1944	139			7%	9
Heat treatment	12 hr for 104 pcs	187.2	139	104	35	Batch	110
Fitting + Lensing	120 secs	648	139			21%	18
Shot blasting	Batch of 50 in 12 mins	5400	139			Batch	60
Quality Inspection	120 secs	648	139			21%	18
Preturning	328 secs	237	139			59%	18
OP 10	700 secs	222	139			63%	18
Washing	80 secs	972	139			14%	9
CNC	40 secs	1944	139			7%	9
Air gauge	120 secs	648	139			21%	18
Pump housing bush press	300 sec	259	139			54%	18
OP 20	664 secs	234	139			59%	18
OP 30	690 secs	225	139			62%	18
Washing 1	60 sec for 2 pcs together	2592	139			5%	9
ECM	72 secs	1080	139			13%	9
Washing 2	30 secs	2592	139			5%	9
LT	210 sec	370	139			38%	18
Multi	270 sec	288	139			48%	18
Endoscopy	180 secs	432	139			32%	18
CSL	300 secs	259	139			54%	18
Ultrasonic Washing	10 units 18 mins	720	139			19%	18
Packing	60 sec	1296	139			11%	18
Total WIP							506

Borgwarner EAD HP

Table 3.7 Cycle time information of Borgwarner EAD HP

Steps	Cycle time
Melting	1 ton 1hr
Coring	160 sec
GDC	140 sec
Decoring	60 sec
Gate cutting	60 sec
Heat Treatment (T5)	7.5 hr for 300 batch
Knocking	30 sec
Shot blasting	450 sec for 100 pcs
Sand blasting	5 sec
Rework	60 sec
Inspection	60 sec
OP 10	213 sec
OP 20	275 sec
OP 30/40	290 sec
OP 50	295 sec
Deburring	90 sec
Water dip	90 sec
Washing	120 sec
DNC	90 sec
Final Endoscopy	120 sec
CAP Assembly	90 sec
Multigaging	90 sec
LT 1	90 sec
LT 2	120 sec
CSL	120 sec
Packaging	60 sec

3.9 PROCESS FLOW

Studied the end-to-end process flow of Borgwarner EAD HP and monitored the steps involved in the production process to make an inventory plan.

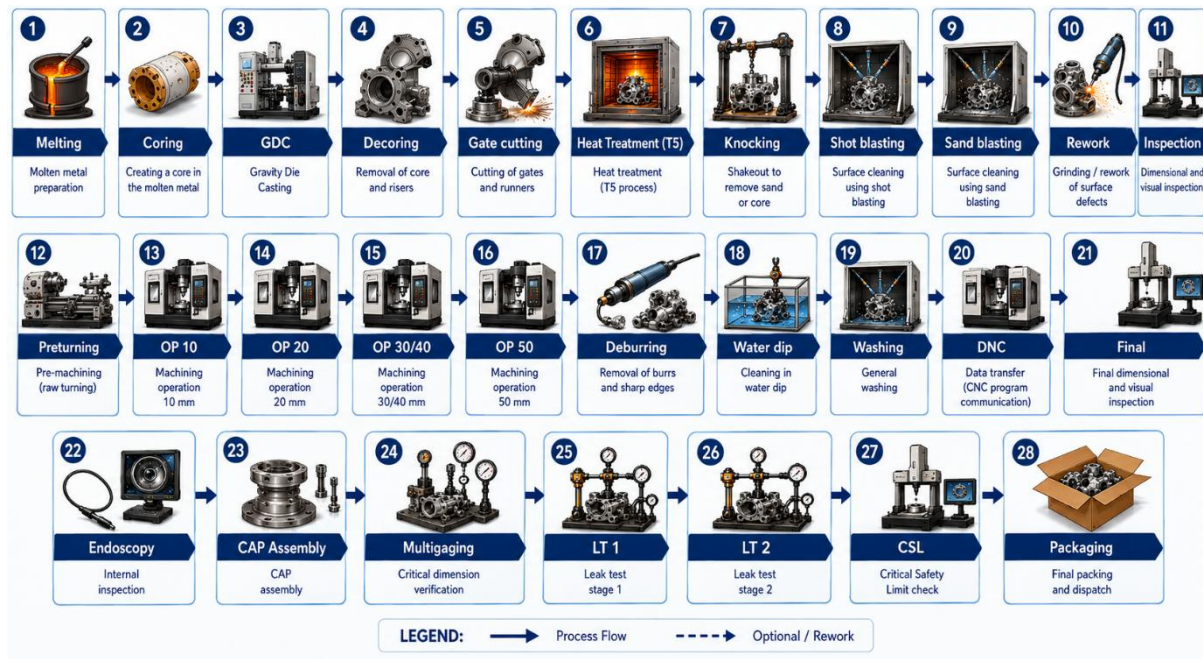


Fig 3.3 Production process flow of Borgwarner EAD HP

3.10 LEAD TIME ANALYSIS

A lead time analysis was conducted for the Borgwarner EAD HP manufacturing line to understand the material flow from initial melting to the final finished good. The analysis was done under initial production startup conditions, assuming no WIP is present in stages. The analysis aimed to identify the bottlenecks, the accumulation delays and their impact on overall lead time.

Once steady-state production is achieved, material flow becomes continuous, and downstream operations receive regular inputs from the upstream process.

The cumulative lead time was calculated by analyzing the cycle time, batch size and accumulation delays between the processes.

Table 3.8 Lead Time Analysis of Borgwarner EAD HP

Steps	Cycle time	Explanation	Lead Time in hr
Melting	1 ton 1hr	1.173 kg is the raw weight and 600 gm is final weight 1ton can produce 908 castings	1.00
Coring	160 sec		2.10
GDC	140 sec	26 machines are available and 25 castings can be done per hour	8.10
Decoring	60 sec	0.41 hours for 25 castings	8.51
Gate cutting	60 sec	0.41 hours for 25 castings	8.91
Heat Treatment (T5)	7.5 hr for 300 batch	300 pcs can be done in 1 batch of 7.5 hrs but in this case we do only 231 but the time remains same. First lot of 25 castings has to wait for 9.9 hours to get a batch of 231 castings	26.31
Knocking	30 sec	All the 231 units pass continuously and simultaneously through these steps	27.11
Shot blasting	450 sec for 100 pcs		27.24
Sand blasting	5 sec		27.24
Rework	60 sec		27.25
Inspection	60 sec		27.27
OP 10	213 sec		27.32
OP 20	275 sec		27.39
OP 30/40	290 sec		27.47
OP 50	295 sec		27.55
Deburring	90 sec		27.57
Water dip	90 sec		27.60
Washing	120 sec		27.63
DNC	90 sec		27.66
Final Endoscopy	120 sec		27.69
CAP Assembly	90 sec		27.71
Multigaging	90 sec		27.74
LT 1	90 sec		27.76
LT 2	120 sec		27.79
CSL	120 sec		27.82
Packaging	60 sec		27.84

3.11 SAFETY STOCK CALCULATION AND INVENTORY PLANNING

The monthly demand for Borgwarner EAD HP from the client is 6000 units

The number of working days in the plant is 26 days

Total hours, including all shifts per day, is 24 hrs and 86400 secs

Based on this, the daily production is 231 units per day (6000/26)

Hourly Production rate is 10 units/hr (231/24)

TAKT time is 374.02 secs (86400/231)

The safety stock factor we have considered for this was 1.5

The safety stock formula used is

Minimum Safety stock = Recovery time * Safety stock factor * Demand rate per hour

To determine recovery time, calculate the machines' capacity utilization. The capacity utilization is based on the daily production demand of 231 units/day and does not include safety stock processing. This assumption is intentional because if capacity utilization were calculated including safety stock, it would overstate the load, especially when the production reaches a steady state, and distort the recovery time classification for the machines.

The table shows the amount of Safety stock to be kept in each stage so that machine breakdown or other delays don't stop the production process, and the process is steady and continues uninterrupted.

Recovery Time Assumption

For machines with more than 20% capacity utilization, we have considered a 2hr recovery time.

For machines with less than 20% capacity utilization, we have considered a 1 hr recover as these are fast and can cope with the process delays.

And for the batch process, we will keep one batch and a little extra, considering the delay as a safety stock.

Table 3.9 Safety Stock Analysis of Borgwarner EAD HP

Steps	Cycle time	Capacity	Per Day produce	Parts in WIP	Capacity Utilization	Minimum safety stock
Melting	1 ton 1hr					
Coring	160 sec	540	231		43%	30
GDC	140 sec	617	231	231	37%	29
Decoring	60 sec	1440	231		16%	14
Gate cutting	60 sec	1440	231		16%	14
Heat Treatment (T5)	7.5 hr for 300 batch	900	231		Batch	80
Knocking	30 sec	2880	231		8%	14
Shot blasting	450 sec for 100 pcs	19199	231		1%	14
Sand blasting	5 sec	17280	231		1%	14
Rework	60 sec	1440	231		16%	14
Inspection	60 sec	1440	231		16%	14
OP 10	213 sec	405	231		57%	29
OP 20	275 sec	314	231		74%	29
OP 30/40	290 sec	297	231		78%	29
OP 50	295 sec	292	231		79%	29
Deburring	90 sec	960	231		24%	29
Water dip	90 sec	960	231		24%	29
Washing	120 sec	720	231		32%	29
DNC	90 sec	960	231		24%	29
Final Endoscopy	120 sec	720	231		32%	29
CAP Assembly	90 sec	960	231		24%	29
Multigaging	90 sec	960	231		24%	29
LT 1	90 sec	960	231		24%	29
LT 2	120 sec	720	231		32%	29
CSL	120 sec	720	231		32%	29
Packaging	60 sec	1440	231		16%	
Total WIP						614

Borgwarner EAD LP

Table 3.10 Cycle time information of Borgwarner EAD LP

Steps	Cycle time
Melting	1 ton 1 hr
Coring	230 sec
GDC	210 sec
Decoring	60 sec
Gatecutting	60 sec
Heat Treatment (T5)	7.5 hr
Knocking	30 sec
Shot Blasting	7.5 min
Sand blasting	5 sec
Rework	60sec
Inspection	60sec
Preturning	90 sec
OP 10	166 sec
OP 20	245 sec
OP 30/40	273 sec
OP 50	258 sec
Deburring	90 sec
Water dip	90 sec
Washing	120 sec
DNC	90 sec
Final Endoscopy	120 sec
CAP Assembly	90 sec
Multigaging	90 sec
Bush Assembly	90 sec
LT 1	90 sec
LT 2	120 sec
CSL	120 sec
Packaging	60 sec

3.12 PROCESS FLOW

Studied the end-to-end process flow of Borgwarner EAD LP and monitored the steps involved in the production process to make an inventory plan.

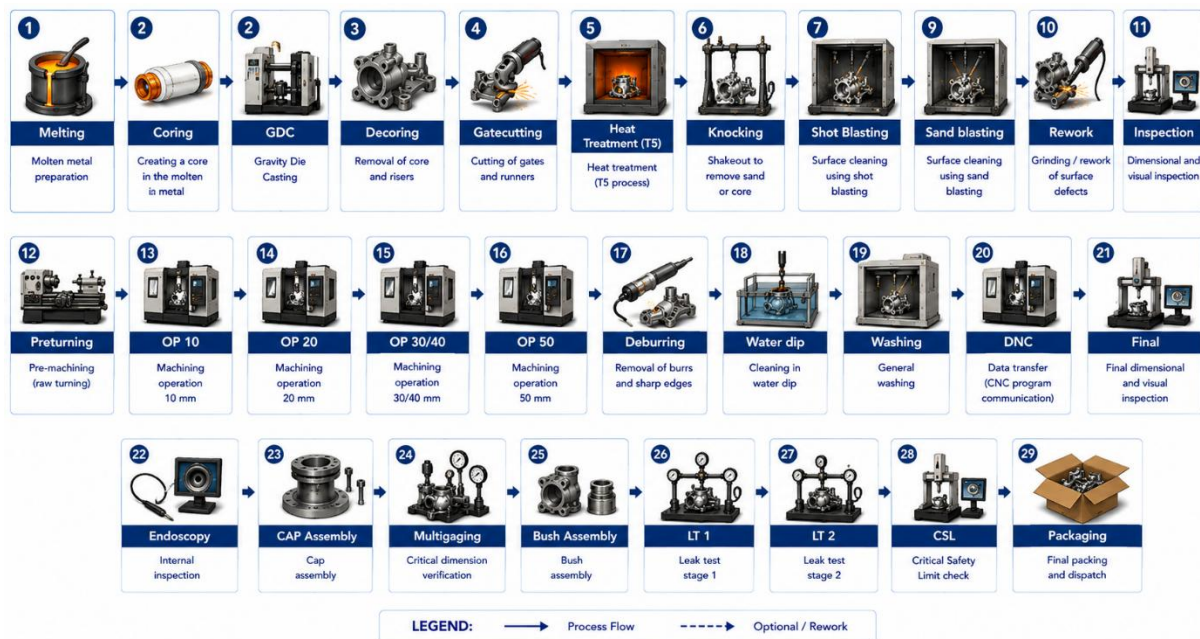


Fig 3.4 Production process flow of Borgwarner EAD LP

3.13 LEAD TIME ANALYSIS

A lead time analysis was conducted for the Borgwarner EAD LP manufacturing line to understand the material flow from initial melting to the final finished good. The analysis was done under initial production startup conditions, assuming no WIP is present in stages. The analysis aimed to identify the bottlenecks, the accumulation delays and their impact on overall lead time.

Once steady-state production is achieved, material flow becomes continuous, and downstream operations receive regular inputs from the upstream process.

The cumulative lead time was calculated by analyzing the cycle time, batch size and accumulation delays between the processes.

Fig 3.11 Lead Time Analysis of Borgwarner EAD LP

Steps	Cycle time	Explanation	Lead time (hrs)
Melting	1 ton 1 hr	2.319 kg is the raw weight final is 1.4 kg can produce 431 casting	1.00
Coring	230 sec		2.08
GDC	210 sec	26 machines are available 17 castings will be done in 1hr	7.08
Decoring	60 sec		7.36
Gatecutting	60 sec		7.64
Heat Treatment (T5)	7.5 hr	300 pcs can be done in 1 batch of 7.5 hrs here we will do with 231 castings and first lot of 17 castings has to wait for 14 hrs	29.14
Knocking	30 sec		29.94
Shot Blasting	7.5 min	100 pcs done in 7.5 mins	30.07
Sand blasting	5 sec	All the 231 units pass continuously and simultaneously through this steps	30.07
Rework	60sec		30.08
Inspection	60sec		30.09
Preturning	90 sec		30.11
OP 10	166 sec		30.16
OP 20	245 sec		30.22
OP 30/40	273 sec		30.29
OP 50	258 sec		30.36
Deburring	90 sec		30.39
Water dip	90 sec		30.41
Washing	120 sec		30.45
DNC	90 sec		30.47
Final Endoscopy	120 sec		30.50
CAP Assembly	90 sec		30.53
Multigaging	90 sec		30.55
Bush Assembly	90 sec		30.58
LT 1	90 sec		30.60
LT 2	120 sec		30.63
CSL	120 sec		30.66
Packaging	60 sec		30.67

3.14 SAFETY STOCK CALCULATION AND INVENTORY PLANNING

The monthly demand for Borgwarner EAD LP from the client is 6000 units

The number of working days in the plant is 26 days

Total hours, including all shifts per day, is 24 hrs and 86400 secs

Based on this, the daily production is 231 units per day (6000/26)

Hourly Production rate is 10 units/hr (231/24)

TAKT time is 374secs (86400/231)

The safety stock factor we have considered for this was 1.5

The safety stock formula used is

Minimum Safety stock = Recovery time * Safety stock factor * Demand rate per hour

To determine recovery time, calculated the machines' capacity utilization. The capacity utilization is based on the daily production demand of 231 units/day and does not include safety stock processing. This assumption is intentional because if capacity utilization were calculated including safety stock, it would overstate the load, especially when the production reaches a steady state, and distort the recovery time classification for the machines.

The table shows the amount of Safety stock to be kept in each stage so that machine breakdown or other delays don't stop the production process, and the process is steady and continues uninterrupted.

Recovery Time Assumption

For machines with more than 20% capacity utilization, we have considered a 2hr recovery time.

For machines with less than 20% capacity utilization, we have considered a 1 hr recover as these are fast and can cope with the process delays.

And for the batch process, we will keep one batch and a little extra, considering the delay as a safety stock.

Fig 3.12 Safety Stock Analysis of Borgwarner EAD LP

Steps	Cycle time	Capacity	Per Day produce	Parts in WIP	Capacity Utilization	Minimum safety stock
Melting	1 ton 1 hr					
Coring	230 sec	375	231		62%	30
GDC	210 sec	411	231	231	56%	29
Decoring	60 sec	1440	231		16%	14
Gatecutting	60 sec	1440	231		16%	14
Heat Treatment (T5)	7.5 hr	900	231		26%	80
Knocking	30 sec	2880	231		8%	14
Shot Blasting	450 sec for 100pcs	19199	231		1%	14
Sand blasting	5 sec	17280	231		1%	14
Rework	60sec	1440	231		16%	14
Inspection	60sec	1440	231		16%	14
Preturning	90 sec	960	231		24%	29
OP 10	166 sec	520	231		44%	29
OP 20	245 sec	353	231		66%	29
OP 30/40	273 sec	316	231		73%	29
OP 50	258 sec	335	231		69%	29
Deburring	90 sec	960	231		24%	29
Water dip	90 sec	960	231		24%	29
Washing	120 sec	720	231		32%	29
DNC	90 sec	960	231		24%	29
Final Endoscopy	120 sec	720	231		32%	29
CAP Assembly	90 sec	960	231		24%	29
Multigaging	90 sec	960	231		24%	29
Bush Assembly	90 sec	960	231		24%	29
LT 1	90 sec	960	231		24%	29
LT 2	120 sec	720	231		32%	29
CSL	120 sec	720	231		32%	29
Packaging	60 sec	1440	231		16%	
Total WIP						671

3.15 RESULT AND DISCUSSIONS

Summary of Key Findings and Strategy Implemented

Table 3.13 Summary of Key Findings

Component	Primary Bottleneck	Lead Time in the startup condition	Impact on production flow	Control strategy
Porsche Housing RH	T6 Heat treatment	44.7 hrs	Material accumulation and increased lead time	Drum buffer rope scheduling with strategic inventory
Phinia 6 cylinder	OP 10	29.8 hrs	Limited impact because of parallel setup	Controlled WIP and monitoring of machine operations
Borgwarner EAD HP and EAD LP	OP 30/40 and OP 50	27.84 hrs and 30.67 hrs	Higher cycle time than other processes involved, causing temporary inventory accumulation	Controlled WIP movement and TOC-based inventory planning

The study was conducted to develop an optimized inventory flow for Porsche Housing RH, Phinia 6-cylinder, Borgwarner EAD LP and EAD HP through cycle time analysis, bottleneck assessment, and safety stock planning. The analysis revealed that although the manufacturing process differed across products, each production flow contained some bottlenecks, which increased the lead time of the production.

For Porsche housing RH, the cumulative lead time was estimated at 44.71 hrs under startup conditions. The analysis showed that the T6 heat treatment has the highest 93% capacity utilization, making it the primary bottleneck of the production process. The batch process nature of T5 and shot blasting operations contributed to the significant increase in the lead time and reduced flow efficiency.

In the Phinia 6 cylinder production line, OP 10 was identified as the primary bottleneck, followed by OP 20 and OP 30. These operations possessed the highest machining cycle times within the process and, therefore, had the potential to restrict throughput. However, the availability of parallel machines reduced the severity of these bottlenecks and enabled the production system to maintain the output rate.

The Borgwarner EAD LP and EAD HP revealed that the primary constraints within both production flows were mostly associated with high machine utilization. The interaction between casting, heat treatment, machining, inspection and assembly operations created temporary inventory buildup at specific points in the process. The lead time analyses demonstrated that the waiting time before these constrained operations contributed significantly more to the total manufacturing lead time than the actual processing time of several activities. These findings highlight the importance of synchronizing production flows around system constraints and critical operations.

Based on the identified bottlenecks and production demand requirements, safety stock levels were calculated for each stage using a recovery time-based assumption. Higher safety stock buffers were recommended for operations exhibiting high utilization and longer recovery periods, while low buffers were recommended for machines with low utilization and substantial excess capacity. The proposed safety stock framework protects against machine breakdown and short-term disruption in production flow, making it more optimized and efficient.

The Drum buffer rope methodology was also applied to synchronize the production flow as per the bottleneck capacity to prevent the generation of excess unwanted WIP in the process, resulting in a leaner process.

Overall, the findings demonstrated that bottlenecks and the batch processing operations were the primary contributors to lead time and inventory accumulation across all four production processes. The integration of cycle time analysis, lead time assessment, safety stock planning and the theory of constraints principles facilitated the development of a structured inventory management framework capable of improving material flow, reducing production interruptions and enhancing efficiency in the die casting manufacturing environment.

CHAPTER 4

PROJECT 2

4.1 OBJECTIVE

Developing an Overall Equipment Effectiveness (OEE) tracking and monitoring sheet based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines so that data-driven decisions can be taken and appropriate future production planning can take place, helping the plant to achieve the production targets.

4.2 BACKGROUND KNOWLEDGE

GDC (Gravity Die Casting)

The gravity die casting (GDC) method is a process of creating castings from molten metals. The process of GDC does not make use of any pressure or force other than that of gravity to fill permanent metallic dies with molten metals to form castings.

Stationery GDC Process

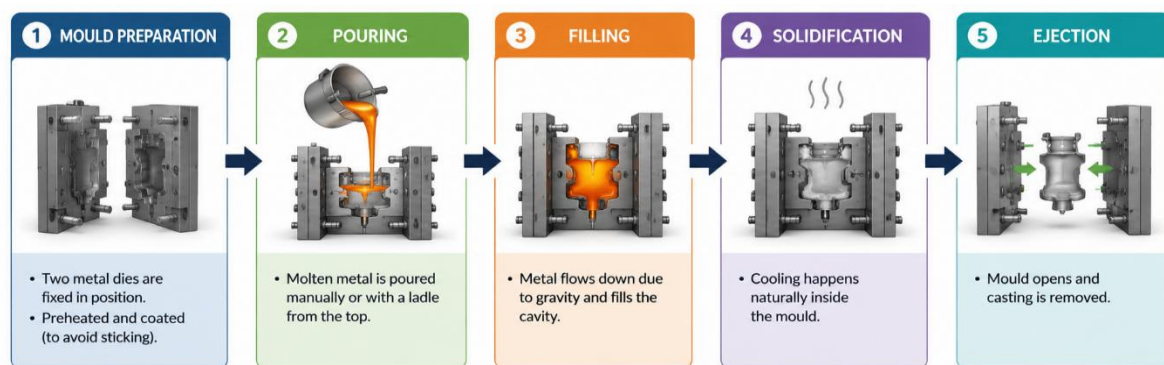


Fig 4.1 Process flow of Stationery GDC

Tilting GDC Machine Process

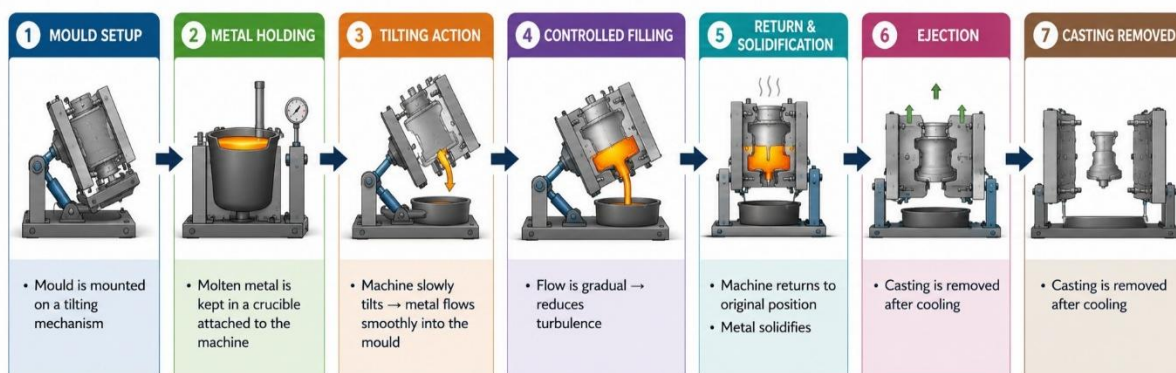


Fig 4.2 Process flow of Tilting GDC

Benefits of GDC

- GDC castings have a much smoother surface finish than sand castings.
- GDC castings have a much better dimensional accuracy than sand castings.

Limitations of GDC

- GDC is slower than pressure die casting.
- GDC does not yield castings with complex thin-wall shapes.

In the plant, there are four different GDC lines, which are used for making GDC castings

- Tilting GDC line
- Diamler Robo line
- Tilting Robo line
- Stationery Line

LPDC (Low Pressure Die Casting)

Low-pressure die casting (LPDC) is a process where molten metal is pushed into the mould using low air pressure.

Process

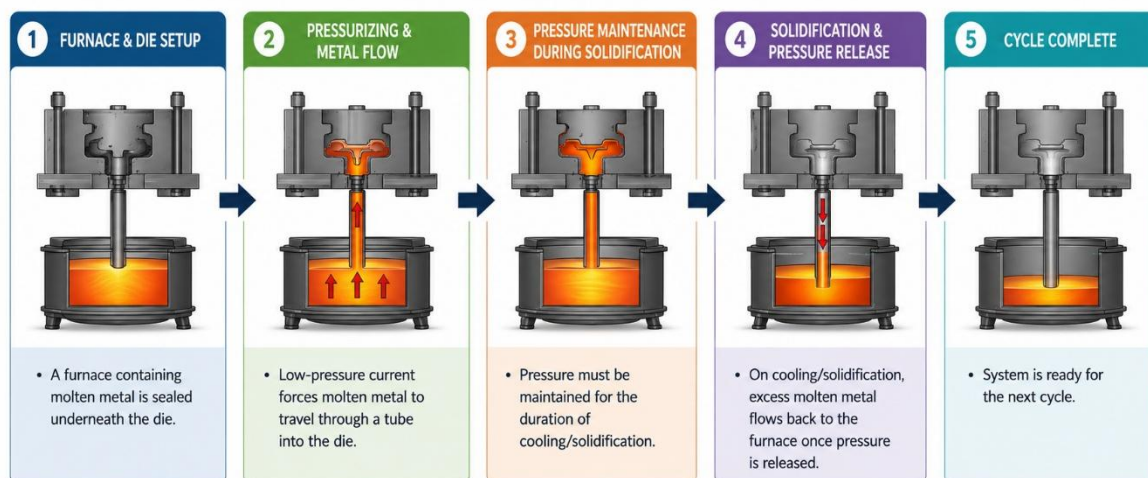


Fig 4.3 Process flow of LPDC

Advantages of LPDC

- Very low porosity
- Increased strength
- Superior surface quality

Limitations of LPDC

- Higher initial cost of equipment
- Longer cycle time compared to HPDC
- Generally, applicable to non-ferrous metals only

4.3 METHODOLOGY

- Received the daily data of the LPDC and GDC machines, which had the daily breakdown details and efficiency details of both machine lines
- Checked the data for data inconsistencies, performed data cleaning and verified the data with the respective supervisors and operators to ensure the data used is correct
- Organized the daily data sheets in Excel and created a new sheet which would contain all the weekly data for the machines. Used the appropriate formula to get the weekly summed-up values of breakdown points, and also calculated the weekly total efficiency
- Used charts to compare the weekly efficiency of different machines present

4.4 RESULTS AND DISCUSSIONS

- This sheet helped the plant head and other executives to detect which are the major breakdown points and downtime causes affecting the production process of both GDC and LPDC machines. This allowed the production and maintenance teams to focus on critical areas requiring immediate corrective actions
- The implementation of the OEE tracking system also supported data-driven decision-making for future production planning. The availability of accurate daily performance data enabled better estimation of machine capacity, improved scheduling of preventive maintenance activities, and reduction of unexpected production interruptions.

CHAPTER 5

CONCLUSION

The Internship at Spark Minda's die-cast plant encompassed two projects aimed at improving operational efficiency on the shop floor.

Project 1 focused on designing a data-driven inventory flow for four die-cast components: Porsche housing RH, Phinia 6-cylinder, Borgwarner EAD LP, and EAD HP. Through systematic cycle-time analysis, lead-time analysis, and bottleneck identification, the study established a scientific basis for inventory planning across the production stages of these components. Bottlenecks were identified based on the capacity utilization and availability of the machine lines, safety stock was calculated using a recovery time framework and drum buffer rope model was recommended to synchronize the production across bottlenecks. The outcome is a structured plant validated inventory management framework that will help in reducing production interruptions, controlling WIP buildup and improving the material flow.

Project 2 focused on tracking and improving Overall Equipment Effectiveness (OEE) across key machines in the die cast division. A structured OEE monitoring system was developed to capture the performance and efficiency of the GDC and LPDC machines, enabling supervisors to identify the recurring breakdown patterns and efficiency losses. This will help the plant to prioritize maintenance interventions and improve equipment uptime over time.

Together, both projects address increasing the efficiency of the production process, one from an inventory planning perspective and the other from an overall equipment efficiency perspective. The combined impact of this project will lead to increasing the efficiency of production in the plant and help in meeting the client demands consistently and sustainably.

CHAPTER 6

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COURSE MAPPING

Problem Statement	FMDM	MaE	MM-1	OB1	SMDM
Designing an optimized inventory flow for selected die-cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification					
Development of an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines					TRUE

Problem Statement	Business Communication	KY	EEP	CMA	MaE-2
Designing an optimized inventory flow for selected die-cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification			TRUE		
Development of an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines			TRUE		

COURSE MAPPING

Problem Statement	BECGS	POM	BA	OB-II	SM
Designing an optimized inventory flow for selected die-cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification		TRUE			
Development of an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines					

Problem Statement	BRM	FM	OTDM	BI	HRM	CAT
Designing an optimized inventory flow for selected die-cast components of Phinia 6 cylinder, Porsche housing RH and Borgwarner EAD HP and EAD LP through cycle time analysis and bottleneck identification						
Development of an Overall Equipment Effectiveness (OEE) tracking and monitoring model based on the daily data to study and analyze the different breakdown points of the GDC and LPDC machines				TRUE		

